

Agentic artificial intelligence in food science: From automation to adaptation

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ABSTRACT

This review synthesizes recent research on the emergence of agentic artificial intelligence in food, agricultural, and nutrition systems. While AI tools are increasingly deployed across domains from crop monitoring and processing to personalized nutrition and supply-chain optimization, current systems remain predominantly task-based, predictive, and narrow in scope. Recent scholarship highlights limitations in contextual reasoning, multi-objective decision-making, and the integration of ethical or governance constraints. This review organizes the literature across agricultural production, food processing, nutrition and health, food safety, and sustainability governance, and assesses how emerging agentic AI architectures may address existing shortcomings. Illustrative examples from food fermentation and personalized nutrition demonstrate persistent challenges when AI systems attempt to operate in real-world contexts characterized by social, ecological, and institutional complexity. The review concludes with implications for transparency, interoperability, and responsible governance, emphasizing the need for AI systems capable of deliberation, adaptive reasoning, and alignment with human and societal values.

1. Introduction

Artificial intelligence is increasingly embedded across food, agricultural, and nutrition systems, where it supports prediction, detection, optimization, and process automation. Recent studies have highlighted both the transformative potential of these tools and persistent limitations in their capacity to reason across ecological, social, ethical, and operational trade-offs (Birhane, 2021; Van Der Burg et al., 2024). Rather than offering a new commentary, this review synthesizes emerging research on how AI systems are currently used in food systems, where they fall short, and how agentic AI architectures may address these challenges. By bringing together insights from agriculture, processing, nutrition, and sustainability governance, the review identifies cross-cutting themes and areas requiring further evidence, methodological development, and policy attention.

AI applications in food science have historically drawn from three methodological families: rule-based expert systems for decision support, machine-learning approaches such as random forests and SVMs for prediction and classification, and deep-learning architectures including CNNs and transformers for image, sensor, and molecular data analysis. These approaches form the technical foundation from which more

adaptive and agentic architectures are now emerging (Onyeaka et al., 2025).

The idea of **agency** challenges this paradigm. *Agentic AI* refers to intelligent systems capable of autonomous reasoning and adaptation, not in isolation from humans but in coordination with these agents (Zou & Topol, 2025). Such systems can pursue dynamic goals balancing productivity with sustainability, or food safety with affordability by continuously integrating feedback from the environments they influence.

This review identifies that adopting an agentic perspective requires more than just technical refinement; it demands rethinking of how intelligence itself is conceptualized within food systems. The following sections treat the deployment of agentic intelligence from conceptual definition, to architectural design, to concrete applications and governance implications.

2. Review methods

This article is structured as a narrative literature review synthesizing peer-reviewed research published between 2018 and 2025 on artificial intelligence in food, agricultural, nutrition, and sustainability domains.

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Relevant literature was identified through targeted searches in Scopus, Web of Science, and Google Scholar using combinations of terms such as “agentic AI,” “AI in food systems,” “digital agriculture,” “context-aware AI,” “AI governance,” “food processing automation,” and “personalized nutrition AI.” Additional publications were located through backward and forward citation tracking of key papers. Because the aim of this review is interpretive rather than exhaustive, no formal inclusion/exclusion criteria or quantitative synthesis procedures were applied. Instead, studies were selected for their conceptual, empirical, or methodological relevance to themes emerging across food production, processing, nutrition, safety, and governance. The review integrates these domains to examine how contemporary AI systems operate, where their limitations persist, and how agentic architectures have been proposed to bridge gaps in reasoning, adaptability, and system-level alignment.

3. From task-based AI to agentic intelligence

3.1. Conceptual shift

Task-based AI performs well in static conditions, for example, when predicting crop yield from soil data or detecting contaminants from images. However, when objectives conflict, such as maximizing yield while minimizing carbon intensity, static models fail to reason across competing constraints.

Agentic AI introduces three interdependent capacities (Zhang & Pope, 2025):

- 1. **Goal orientation** – ability to formulate intermediate objectives aligned with overarching aims.
- 2. **Contextual reasoning** – capacity to interpret uncertainty, negotiate trade-offs, and justify actions.
- 3. **Feedback integration** – continuous learning from environmental and human responses.

Operationally, agentic AI refers to systems that maintain persistent goals, perform multi-step reasoning, adapt plans through feedback loops, and coordinate with humans or other agents in real time. This distinguishes agentic systems from adaptive models, which adjust outputs but do not reformulate objectives or negotiate trade-offs. These properties are described in the literature as redefining automation. Whereas classical AI executes commands, agentic AI can negotiate intentions. Such negotiation is essential in food systems, where data are heterogeneous, incentives plural, and ethics non-negotiable.

3.2. Comparative dimensions

Table 1 summarizes how this shift reshapes not only data usage but epistemic stance: from optimization to adaptation, from prediction to reasoning, and from isolated performance metrics to participatory accountability.

Yet the comparison also exposes the immaturity of the field. Most purportedly “agentic” demonstrations remain pilot-scale, lacking the empirical grounding necessary to establish trust. This raises a central tension that runs through the rest of the paper: can agency be engineered, or must it be cultivated through institutional and cultural change in digital agriculture initiatives? Global digital agriculture initiatives illustrate this tension. While large-scale platforms such as the CGIAR Digital Innovation Hub demonstrate the potential for coordinated data ecosystems, their deployment has also generated debate around data sovereignty, dependency, and governance structures in the Global South (Dittmer et al., 2025; Seid, 2025). These dynamics suggest that technical agency cannot be separated from institutional capacity, political context, and long-term stewardship. As illustrated in Fig. 1, the progression from autonomous to agentic systems highlights how reasoning and planning become central to adaptive food systems intelligence.

Table 1
Comparison of task-based and agentic AI in food systems.

Dimension	Task-based AI	Agentic AI
Goal Orientation	Executes predefined tasks (e.g., classify food, predict yield)	Pursues adaptive goals over time (e.g., optimize diet for health & sustainability simultaneously)
Reasoning Flexibility	Fixed logic or shallow pattern recognition	Capable of multi-step reasoning and trade-off management using techniques like ReAct/ToT
Data Integration	Operates on narrow, static datasets	Integrates multimodal, real-time data (e.g., user preferences, climate, dynamic food prices)
Personalization	Limited to static profiles or single inputs (e.g., calories, allergies)	Dynamically adapts to user behavior, health feedback, and social context
Interactivity	Often unidirectional (e.g., generates one-off recommendations)	Engages in ongoing, conversational, and iterative interaction with users
Tool Use	Rare or manual API integration	Actively uses APIs, external tools, and real-world databases to inform decisions
Adaptability Over Time	Requires retraining or manual updates	Learns from feedback, adapts autonomously within safe boundaries
Human-AI Teaming	Human-in-the-loop typically post hoc (e.g., review outputs)	Designed for collaborative co-decision-making with farmers, consumers, or regulators

4. Architecture and interoperability

Agentic AI in food science can draw on protocols like the Model Context Protocol (MCP) for modular communication among components, enabling reasoning and actuation in wireless-aware environments, each mediated by human oversight (Liu & Du, 2025).

4.1. Reasoning layer

Large-language-model interfaces interpret queries and contextual data from molecular properties to climatic variables. They perform retrieval-augmented reasoning across heterogeneous sources, synthesizing evidence rather than executing fixed algorithms.

4.2. Planning and memory layer

This layer converts reasoning outputs into structured plans, tracks outcomes, and revises goals over time. Here lies the distinction between learning and *meta-learning*: the ability to adapt not only parameters but strategies.

4.3. Actuation layer

Through APIs, IoT networks, or robotic controllers, the actuation layer implements decisions adjusting fermentation parameters, rerouting logistics, or triggering food safety alerts. Note that autonomy emerges from the interplay between these layers, not from any single component.

4.4. Interoperability and infrastructure

The **Model Context Protocol (MCP)** allows communication among independent agents and external databases, offering a lingua franca for reasoning states and tool outputs (Hardt et al., 2017). Compared with industrial data standards such as OPC UA for KPI calculations in manufacturing (Bega et al., 2025) or GS1 (Vlek et al., 2022), MCP extends the notion of interoperability from syntax to semantics. For instance, when integrated with domain ontologies such as FoodOn, MCP

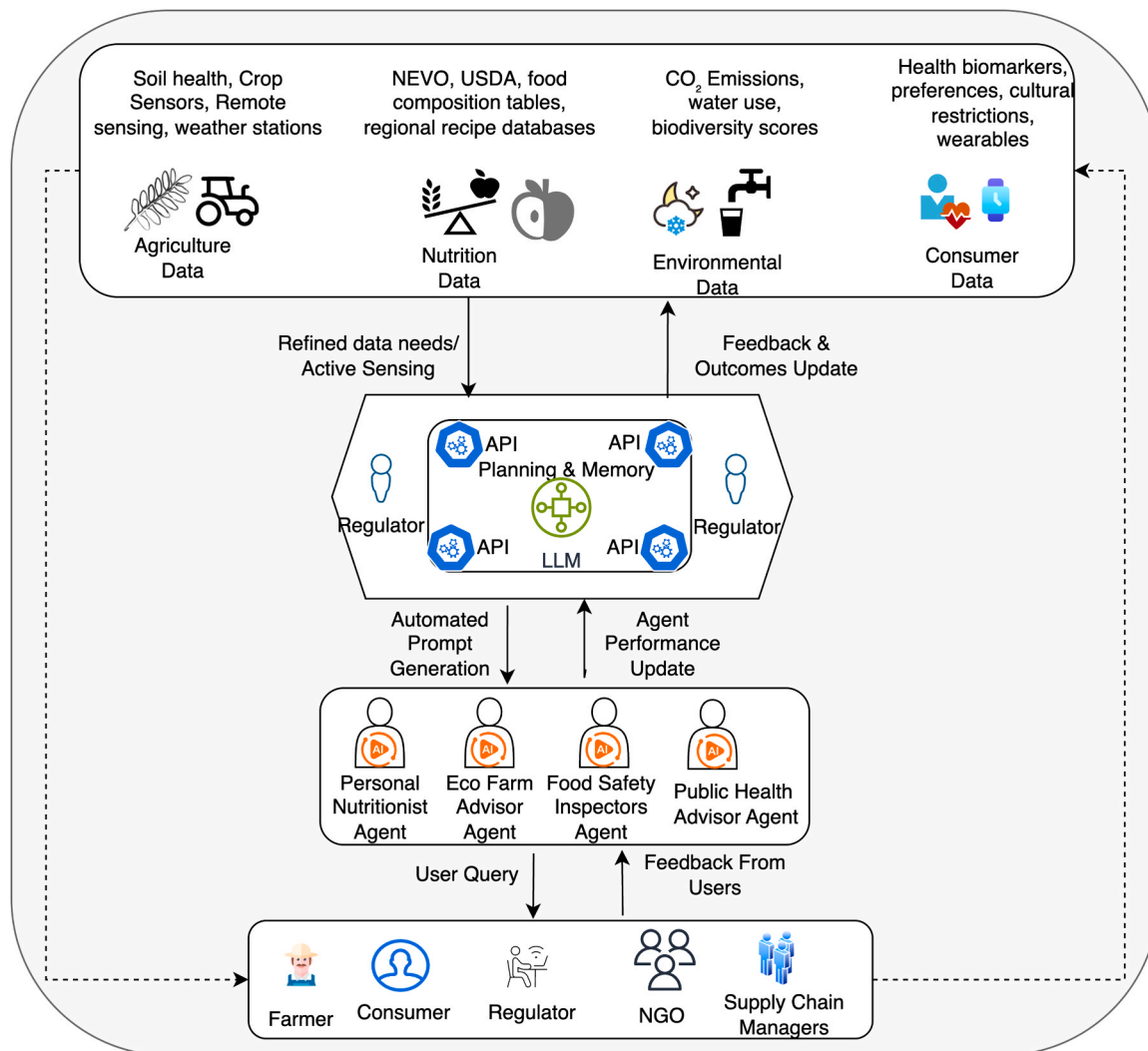


Fig. 1. Progression from autonomous to agentic AI in food systems. Autonomous agents operate in structured environments with reactive, single-task behaviors, while agentic AI agents extend these capabilities through memory, reasoning, and planning to navigate complex, dynamic contexts. The transition toward agentic intelligence enables adaptive task management and proactive interaction with food systems data, supporting continuous engagement across agricultural, processing, and nutritional domains.

can harmonize raw sensor descriptors with standardized food classifications. This enables an agent to interpret heterogeneous signals ranging from soil measurements to nutritional attributes using consistent semantic labels, thereby reducing ambiguity in downstream reasoning. For example, when a sensor reports “moisture_12A” and a vendor dataset reports “soilWaterContent,” MCP resolves the semantic mismatch by mapping both to the standardized ENVO term for “soil water content” (exact synonym: “soil moisture”), which FoodOn also references for environmental attributes, ensuring consistent reasoning across agents. Large models demand infrastructure investments that are unevenly distributed across the Global South (Wu, 2024). Without shared ontologies, open metadata, and reproducible benchmarks, autonomy risks becoming another form of centralization. Fig. 2 summarizes this layered architecture, showing how sensing, reasoning, and planning interact through feedback loops. As shown in Fig. 2, the MCP-mediated interoperability ensures that sensing inputs, reasoning outputs, and planning feedback loops operate cohesively as a unified agentic system. Fig. 2 illustrates the layered architecture of agentic AI in food systems integrating multimodal data sources (agriculture, nutrition, environment, and consumer health). A central reasoning and planning module combines large-language-model (LLM) reasoning with API-based memory and feedback loops. Domain-specific agents

(nutrition, farm, safety, public health) interact with farmers, regulators, and consumers, while regulators provide oversight through feedback and performance updates. The architecture illustrates data reciprocity between sensing, reasoning, and user engagement.

These architectural layers provide the operational foundation for the application domains discussed in Section 5.

5. Applications across the food system

The architecture outlined above gains meaning only when translated into practice. Agentic AI must prove its worth not through conceptual novelty but through its capacity to operate across scales—systemic, processual, and personal. The following subsections trace this continuum from **agriculture** to **nutrition**, showing how agency reframes both the mechanics and the ethics of technological intervention.

5.1. Sustainable agriculture and supply chains

At the system-level, agriculture provides the most visible stage for agentic reasoning. Here, digital technologies can integrate data sources like meteorological forecasts and soil profiles to optimize planting schedules or reroute logistics under climatic stress (Jumanne, 2024).

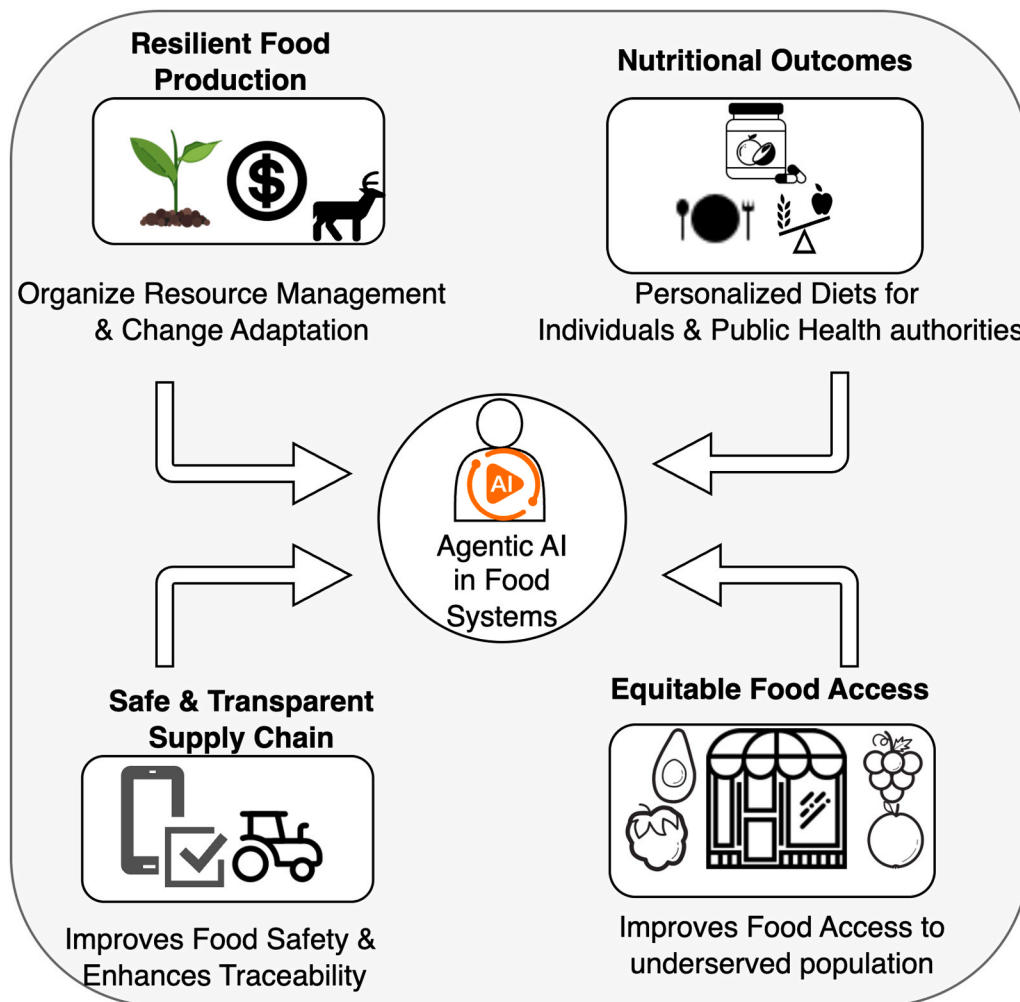


Fig. 2. Layered architecture of agentic AI integrating sensing, reasoning, and planning.

Such integration extends beyond prediction: it enables negotiation among conflicting goals in water management, including equitable access and environmental sustainability. A practical illustration comes from sensor-driven irrigation systems in India, where multi-agent controllers adjust irrigation schedules by simultaneously considering groundwater levels, crop water demand, and household electricity constraints. This deployment constitutes a community-scale field study with limited longitudinal evidence. Field deployments showed water savings of 15–25%, but also revealed governance bottlenecks when local water-user associations lacked decision rights over data access (Singh & Goyal, 2025). It underscores that agentic optimization is only impactful when embedded in social institutions that can act upon model outputs.

Simulations suggest that multi-agent coordination could reduce post-harvest loss by adjusting cold-chain decisions in real time (Shadkam & Irannezhad, 2025). This example is based on simulation evidence rather than empirical deployment. Yet these benefits presuppose data sovereignty and equitable participation. When datasets are proprietary or biased toward industrial production, “autonomy” may simply reinforce existing asymmetries. Thus, the critical question is not *whether* agents can coordinate, but *who* they ultimately coordinate for.

5.2. Food processing and quality control

If agriculture exposes the macro-logic of coordination, food processing tests the micro-logic of precision. Agentic systems embedded in digital-twin factories can calibrate fermentation, drying, or extrusion parameters from continuous sensor feedback, learning to balance safety

thresholds with energy consumption (Ekrami et al., 2025).

However, automation in processing is historically shaped by risk avoidance rather than experimentation. Agentic AI challenges this conservatism: it introduces forms of *controlled improvisation*, where systems explore within regulatory and physical bounds. Whether such improvisation may be tolerated in safety-critical contexts remains to be seen; its success will depend on transparent audit trails and reliable human-override protocols. Parallel advances in intelligent packaging illustrate how AI-enabled sensing and material design are reshaping quality monitoring within food processing systems (Mobahi et al., 2025).

Case. Example: Digital Twins in Cocoa Fermentation (Ghana): A pilot digital-twin fermentation system tested in Ghana integrated temperature sensors with agent-based reasoning to regulate fermentation phases. While the system improved flavor consistency, adoption stalled because community cooperatives lacked control over cloud-stored fermentation data. This illustrates how agentic AI can technically enhance quality yet fail institutionally when governance and data rights are unresolved. This example represents a pilot-scale demonstration rather than a longitudinal field trial, and its evidentiary basis remains preliminary.

5.3. Personalized nutrition and preventive health

At the individual level, personalization becomes the proving ground for ethical agency. Existing recommender systems tailor diets to static

user profiles. Agentic nutrition systems, by contrast, could integrate biomarker data, microbiome shifts, and social context to adapt advice dynamically (Gavai & Van Hillegersberg, 2025).

Yet autonomy in this domain raises a central epistemic tension: does adaptation promote empowerment or drift toward surveillance? The same system that prevents dietary relapse could also nudge behavior beyond the bounds of informed consent. Designing for *bounded agency*, where user feedback constrains algorithmic initiative, may help reconcile personalization with autonomy. Human-centric recommender designs in sustainable food consumption show that semantic modeling can improve transparency and user alignment, supporting agentic co-decision processes (Sedrakyan et al., 2023). Evidence from mobile health interventions reflects this ambiguity. Pilot evaluations of adaptive diet-tracking systems deployed in Brazil and South Africa demonstrated improved adherence in early trials, yet follow-up audits revealed algorithmic nudges that unintentionally penalized users with irregular work schedules or limited food access (Puhl et al., 2022). Such examples show that personalization without contextual sensitivity may amplify rather than reduce health disparities.

Case. Example: Personalized Salt-Reduction Advisor (Japan): Japan’s municipal health agencies piloted an AI-based sodium-reduction advisor that combined diet logs, urine biomarkers, and contextual cues. Although effective in urban users, audits revealed that the system implicitly assumed access to low-sodium food options unavailable in rural regions. This illustrates the risk of agentic personalization drifting toward inequity when models lack embedded knowledge of socioeconomic and geographic constraints. This deployment was evaluated through a municipal pilot program and does not yet have long-term population-level evidence. Thus, the evidence type is early-stage pilot evaluation rather than longitudinal clinical validation.

5.4. Food safety and regulation

Agentic frameworks also reshape the regulatory frontier. Bayesian-network-based vulnerability assessments in spices illustrate how probabilistic reasoning can expose systemic weaknesses in fraud-prone supply chains (Bouzembrak et al., 2024). Recent reviews of AI applications in food safety and quality management also highlight the expanding role of machine-learning pipelines in hazard detection, contamination

assessment, and real-time quality monitoring (Yu et al., 2025). By applying federated learning and ontologies to heterogeneous data, such systems can generate predictions for contamination or fraud (Buyuktepe et al., 2023; Gavai et al., 2023). This aligns with recent advances in AI-driven food adulteration and fraud detection approaches that emphasize combined chemical, spectral, and machine-learning pipelines (Das et al., 2025). But regulatory legitimacy depends on explainability: Black-box reasoning may satisfy engineers but not auditors.

Consequently, agentic systems must externalize their reasoning chains, producing machine-readable justifications akin to “digital dosiers.” Without such transparency, agentic oversight risks becoming another invisible bureaucracy precisely the outcome that safety regimes were designed to prevent.

5.5. Equity and food literacy

Finally, equity defines the moral horizon of agentic transformation. Inclusive digital innovations, including low-bandwidth tools, could democratize access to nutrition education in marginalized regions (Opola et al., 2025). Yet even technologies designed for inclusion can replicate exclusion if local knowledge is reduced to data rather than dialog.

Co-design with communities is therefore not ancillary but constitutive of agency itself. A system that learns *from* rather than *about* its users embodies the participatory ethos required for just food systems. As illustrated in Fig. 3, agentic AI operates across four interconnected domains within the food system.

Agentic architectures support four interconnected outcomes: resilient food production through adaptive resource management; safe and transparent supply chains via enhanced traceability; personalized nutritional outcomes for individuals and public-health authorities; and equitable food access for underserved populations. Together, these domains illustrate how agentic reasoning links technological intelligence with social and environmental goals.

Each of the four domains illustrated in Fig. 3 maps directly to the SDG categories summarized in Table 2. The figure depicts functional capabilities, while Table 2 translates these into measurable societal outcomes.

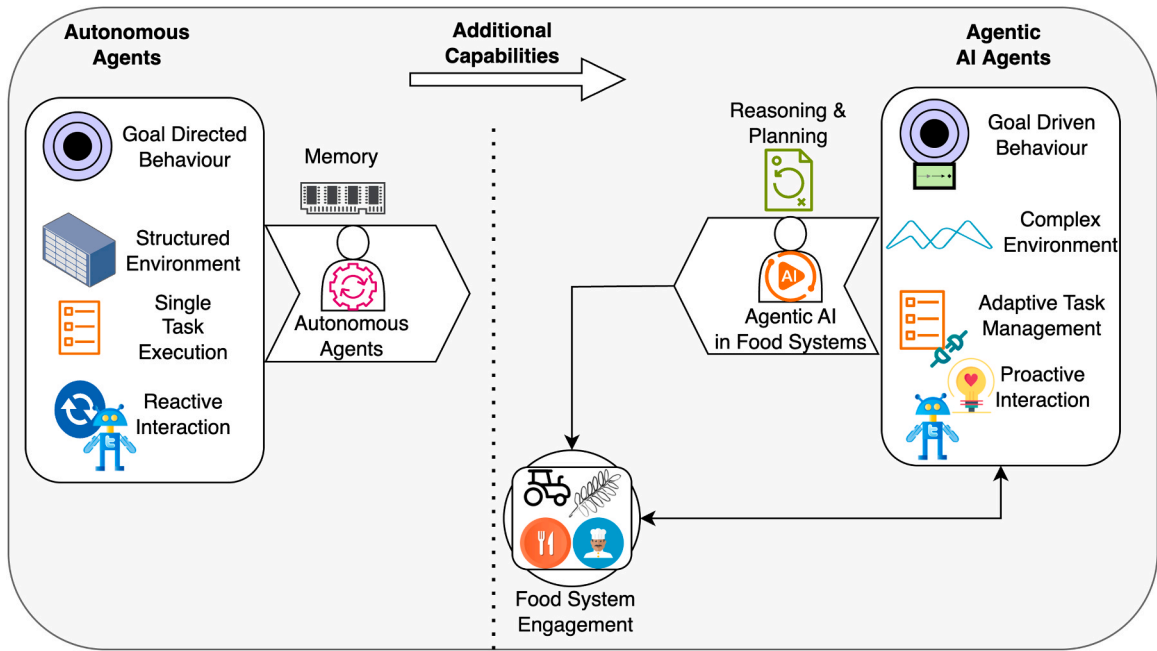


Fig. 3. Application domains of agentic AI.

Table 2
Mapping agentic AI domains to sustainable development goals (SDGs). (Primary SDGs: Core areas directly impacted. Secondary SDGs: Indirectly influenced areas).

Agentic AI domain	Primary SDG contributions	Secondary SDG contributions
Personalized Nutrition & Preventive Health	SDG 3: Good Health and Well-being SDG 12: Responsible Consumption and Production	SDG 13: Climate Action SDG 2: Zero Hunger
Sustainable Agriculture & Supply Chains	SDG 2: Zero Hunger SDG 13: Climate Action	SDG 12: Responsible Consumption and Production SDG 15: Life on Land
Food Safety & Regulation	SDG 12: Responsible Consumption and Production	SDG 3: Good Health and Well-being SDG 9: Industry, Innovation, and Infrastructure
Equity, Access & Food Literacy	SDG 10: Reduced Inequalities SDG 2: Zero Hunger	SDG 4: Quality Education SDG 5: Gender Equality

6. Ethical and governance frameworks

Agentic AI disperses decision-making across human and machine actors, complicating the very notion of accountability. Traditional governance premised on linear causality and *post-hoc* regulation falters when decisions emerge from iterative feedback loops. In this sense, ethics and governance become part of system design rather than an afterthought.

6.1. From compliance to co-evolution

Oversight must shift from static checklists to mechanisms addressing visibility and liability in agent governance (Kolt, 2025). Governance itself thus becomes agentic-responsive, context-aware, and auditable. Key ethical anchors include transparency and tailored explanations, alongside fairness, sustainability, and human oversight; together they form an “ethical quartet”:

- **Transparency** ensures traceability of reasoning chains and data provenance.
- **Fairness** demands bias audits and representative datasets.
- **Sustainability** accounts for energy costs and ecological footprints of model training (Dhayal et al., 2025).
- **Human oversight** preserves interpretive authority and enables contestation of algorithmic outcomes.

These principles parallel foundational frameworks in responsible AI, including the OECD AI Principles (OECD, 2019) the FAO–ITU guidance for AI in agriculture (FAO, 2022), and UNESCO’s Recommendation on the Ethics of Artificial Intelligence (UNESCO, 2021). Incorporating these globally recognized frameworks helps anchor agentic AI within existing governance trajectories rather than treating it as a normative departure. Yet these anchors remain aspirational without institutional scaffolding. Multi-stakeholder governance linking scientists, policymakers, and civil society must evolve from consultation to co-creation. This implies regulatory sandboxes, participatory standard-setting, and global data trusts that safeguard both innovation and equity.

6.2. The problem of trust

Trust cannot simply be legislated; it must be cultivated through iterative transparency. In food systems -where risk perception is visceral- even minor opacity can erode legitimacy. Every act of delegation, from farm sensors to recommendation agents, therefore requires not only technical validation but *narrative accountability* (Gavai &

Bouzembrak, Xhani, et al., 2025). As these systems begin to reason autonomously, governance must shift from controlling outcomes to supervising *processes of reasoning*. The question becomes not “Did the agent comply?” but “How did the agent decide?” Food systems trust differs from medical auditing because it must account for population-level trade-offs among nutrition, ecological constraints, and affordability. This creates what researchers describe as optimization bias, where an agent may prioritize quantifiable efficiencies over social or environmental considerations. In food systems, optimization bias commonly appears when algorithms over-prioritize quantifiable outcomes, such as yield or nutrient density, while under-accounting for affordability or cultural acceptability. Digital dossiers in food safety therefore require audit trails that capture multi-actor decision pathways, heterogeneous data provenance, and contextual justification, rather than the clinician–patient focus common in medical audit trails.

7. Cross-domain lessons

Healthcare offers a cautionary analogy. It demonstrates how validation pipelines, risk classifications, and post-market monitoring can institutionalize algorithmic trust (Karunanayake, 2025). Translating these practices to food science could accelerate acceptance: sandbox testing for food safety algorithms, adaptive certification for evolving models, and registries for algorithmic updates. However, the analogy also reveals structural asymmetries. Medicine benefits from unified vocabularies and rigorous ontologies, while food systems remain fragmented across commodities, regions, and regulatory jurisdictions (Dooley et al., 2018). Data interoperability therefore is not merely technical but political. Commercial confidentiality, trade competition, and divergent cultural conceptions of risk hinder open exchange. A further lesson concerns epistemic humility. Even in medicine -arguably the most data-rich domain- agentic reasoning can amplify existing biases when oversight lags behind innovation. For food science, which encompasses vastly more heterogeneous data, this danger is magnified. Consequently, the challenge is not to replicate healthcare’s regulatory machinery but to adapt its principles -accountability, auditability, and explainability- to the distinct ecology of food systems.

8. Enabling conditions and policy directions

The transition from automation to agency will depend less on technological ingenuity than on the institutions capable of sustaining it. Agentic AI requires an ecosystem that is open, interoperable, and reflexive – qualities still rare in global food data infrastructures. Ultimately, the durability of agentic systems will depend on whether institutions can learn as dynamically as the algorithms they deploy.

8.1. Open infrastructure

A credible foundation demands open frameworks like OpenAg for shared repositories integrating agricultural data (Thudumu & Fisher, 2025). Yet openness is not neutral: It re-distributes power over who defines food knowledge, which languages are codified, and whose experiences remain invisible. Designing open infrastructures therefore means confronting epistemic inequalities as much as engineering protocols.

8.2. Interoperable governance

The European AI Act imposes robustness and cybersecurity requirements (Nolte et al., 2025). Without convergence, agentic initiatives risk fragmenting into compliance silos. An interoperable governance layer should align ethical principles across jurisdictions while allowing contextual flexibility. This demands policy innovation: adaptive

certification, cross-domain sandboxes, and multi-level auditing frameworks that evolve with the systems they supervise.

8.3. Capacity and literacy

Finally, agentic transformation depends on human capability. Food scientists, data engineers, and regulators must learn to interpret reasoning traces, evaluate algorithmic trade-offs, and engage with uncertainty rather than suppress it. Such literacy shifts the question from “Can we trust AI?” to “Are we prepared to participate in its reasoning?” Only then can agency become collective rather than merely delegated.

Investing in these foundations will determine whether agentic AI becomes a practical instrument for sustainability or just another technological narrative.

9. Conclusion

Agentic AI reframes the digital transformation of food science not as a problem of automation but as one of alignment between intelligence, ethics, and planetary limits. It encourages food systems to become adaptive rather than reactive, interpretive rather than merely computational.

Yet agency is not a property of code; it is a relation among actors, infrastructures, and values. Without empirical validation, equitable data access, and reflexive governance, agentic AI risks reproducing the very asymmetries it seeks to overcome. The challenge, then, is not to make machines more autonomous, but to make institutions more accountable for the knowledge such machines produce.

As suggested across the reviewed literature, if automation made food systems efficient, agency may now enable more adaptive and context-sensitive decision-making capable of reasoning across scales, anticipating consequences, and remaining answerable to those whose lives their decisions affect and to the societies and ecologies they shape. Experiences from digital agriculture programs in East Africa and Latin America show that technological sophistication alone does not determine system success; enduring impact arises when tools are co-designed with farmers, regulators, and local institutions. Agentic AI will likely follow a similar trajectory, requiring participatory governance rather than purely technical refinement.

In this sense, the question is not whether food science can build agentic AI, but whether it can build the conditions under which such intelligence remains humane. If successfully realized, agentic AI could transform food systems from reactive infrastructures into adaptive knowledge ecosystems that align productivity, sustainability, and human well-being.

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The author confirms that all content is original, that the manuscript has not been published elsewhere, and that it is not under consideration by another journal. The author approves the final submitted version and agrees to be accountable for all aspects of the work.

CRediT authorship contribution statement

Jaap Heringa: Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **Gavai Anand:** Writing – review & editing, Writing – original draft, Visualization, Methodology,

Conceptualization, Investigation, Validation.

Declaration of Generative AI and AI-assisted technologies in the writing process

During preparation, the authors used *ChatGPT (GPT-5)* for language clarity and section-flow refinement in accordance with Elsevier’s AI-use policy. The authors reviewed and edited all content and take full responsibility for the final manuscript. Figs. 1–3 were created by the authors using standard graphical software tools draw.io (diagrams.net). No generative AI tools were used in the creation or modification of the figures. All figures are original and do not infringe upon any third-party copyrights.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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